```
// Moore model FSM (see Fig. 5.19)                     Verilog 2001, 2005 syntax
module Moore_Model_Fig_5_19 (output [1: 0]  y_out,   input x_in, clock, reset
);
 reg [1: 0] state;
 parameter        S0 = 2'b00, S1 = 2'b01, S2 = 2'b10, S3 = 2'b11;
 always @ (posedge clock, negedge reset)
  if (reset == 0) state <= S0;                          // Initialize to state S0
  else case (state)
   S0:    if (!x_in) state <= S1;    else state <= S0;
   S1:    if ( x_in) state <= S2;    else state <= S3;
   S2:    if (!x_in) state <= S3;    else state <= S2;
   S3:    if (!x_in) state <= S0;    else state <= S3;
  endcase
 assign y_out = state;       // Output of flip-flops
endmodule
```